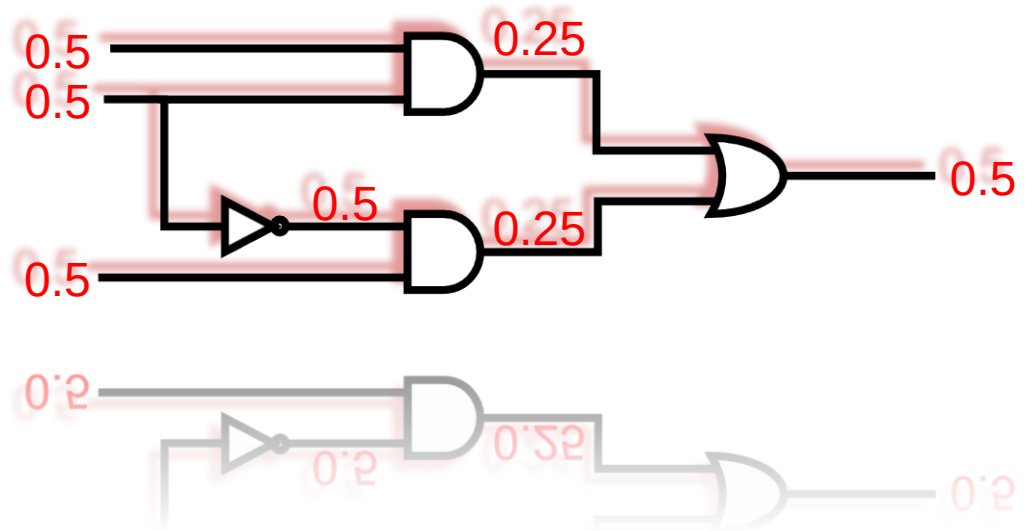


Testability Measure

- Introduction
- SCOAP
- COP
- High-level testability measures
 - ♦ (NOT in EXAM)

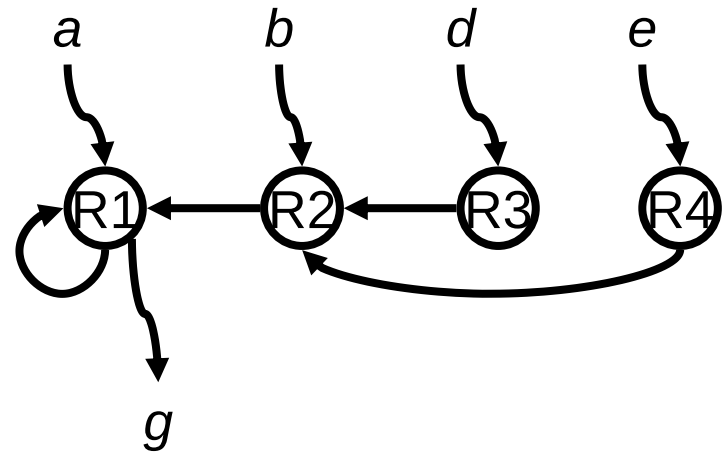
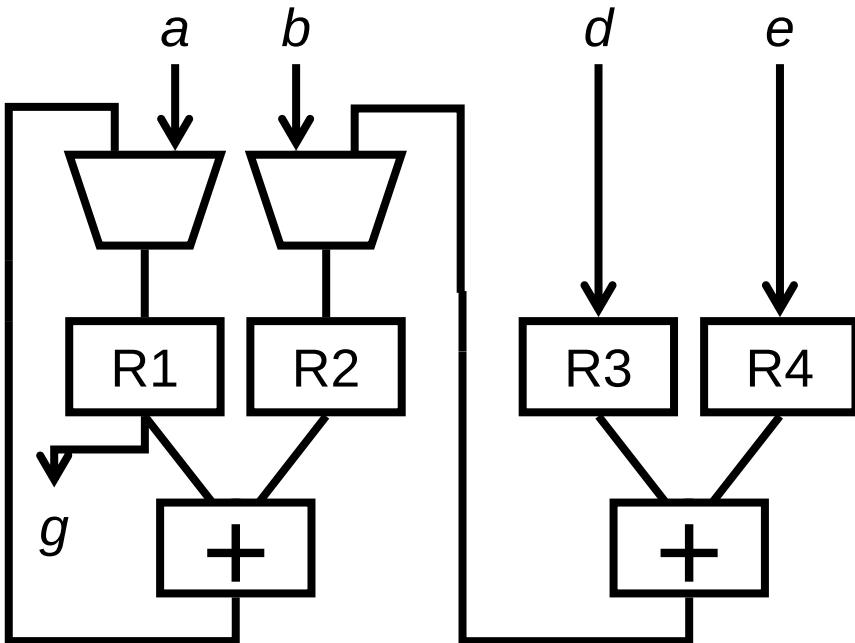


High-Level Testability Analysis

- Based on behavioral level circuit model.
- Usually part of the behavior synthesis program.
 - ♦ To improve the testability at earlier design stage.

Data Flow Graph (DFG)

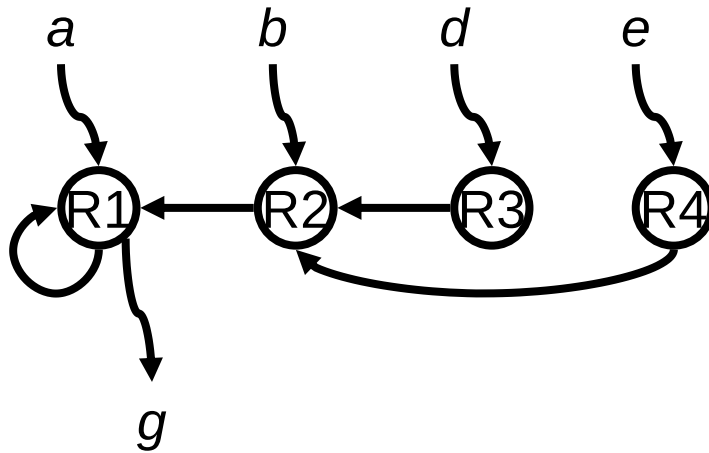
- Each node corresponds to a register.
- Each arc represents a combinational path between two registers.



A High-Level Testability Measure

– Sequential Depth

- The length of a sequential path between two registers is the number of arcs along the path.
- The sequential depth between a pair of registers is the length of the shortest path between them.



$R1 \rightarrow R1 : 0$

$R2 \rightarrow R1 : 1$

$R3 \rightarrow R1 : 2$

$R4 \rightarrow R1 : 2$

$a \rightarrow g : 2$

$b \rightarrow g : 3$

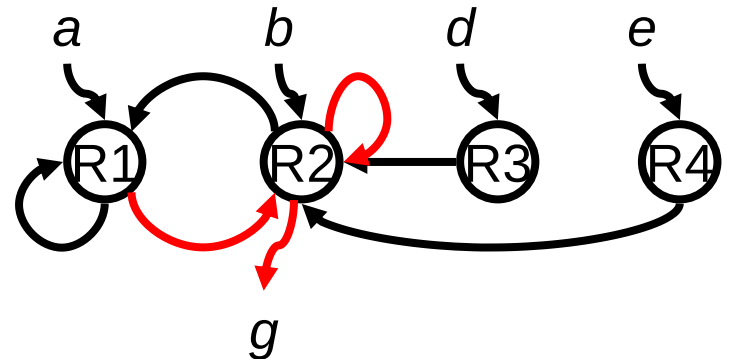
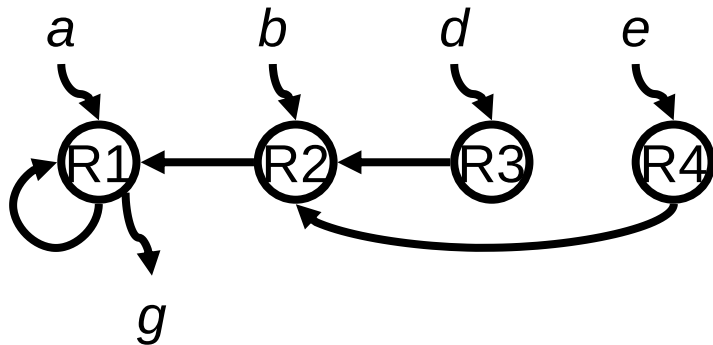
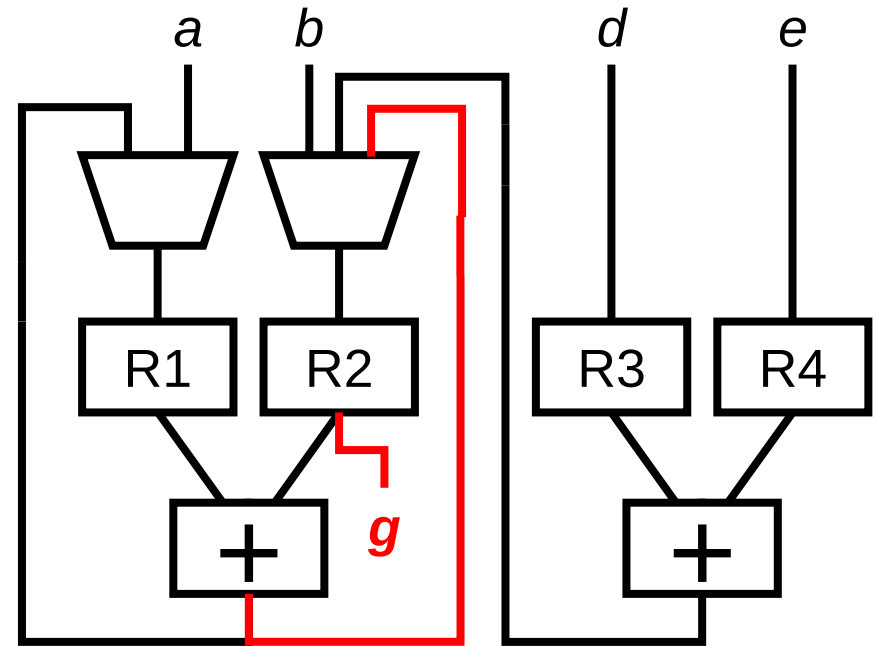
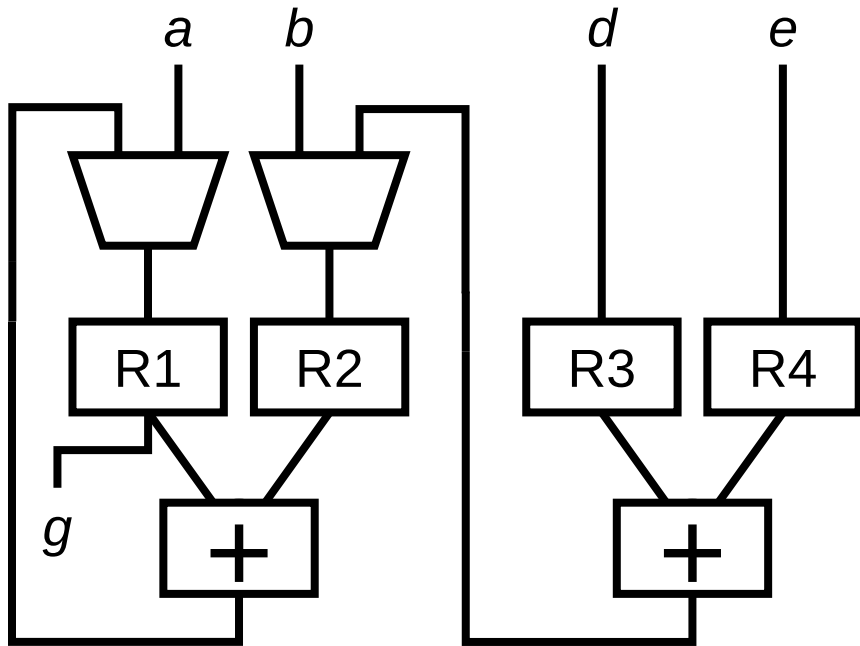
$d \rightarrow g : 4$

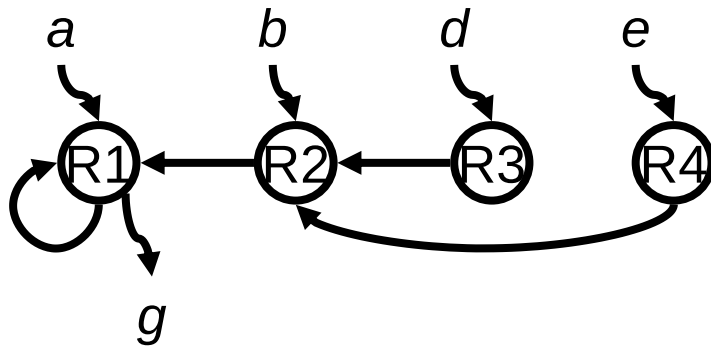
$e \rightarrow g : 4$

Testability Enhancement

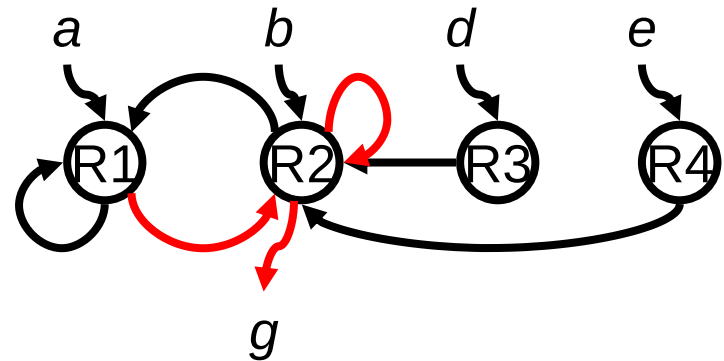
- Improve controllability and observability of registers.
 - ♦ Whenever possible, allocate a register to at least one PI or PO.
- Reduce the sequence depth between a controllable and an observable registers.

An Example

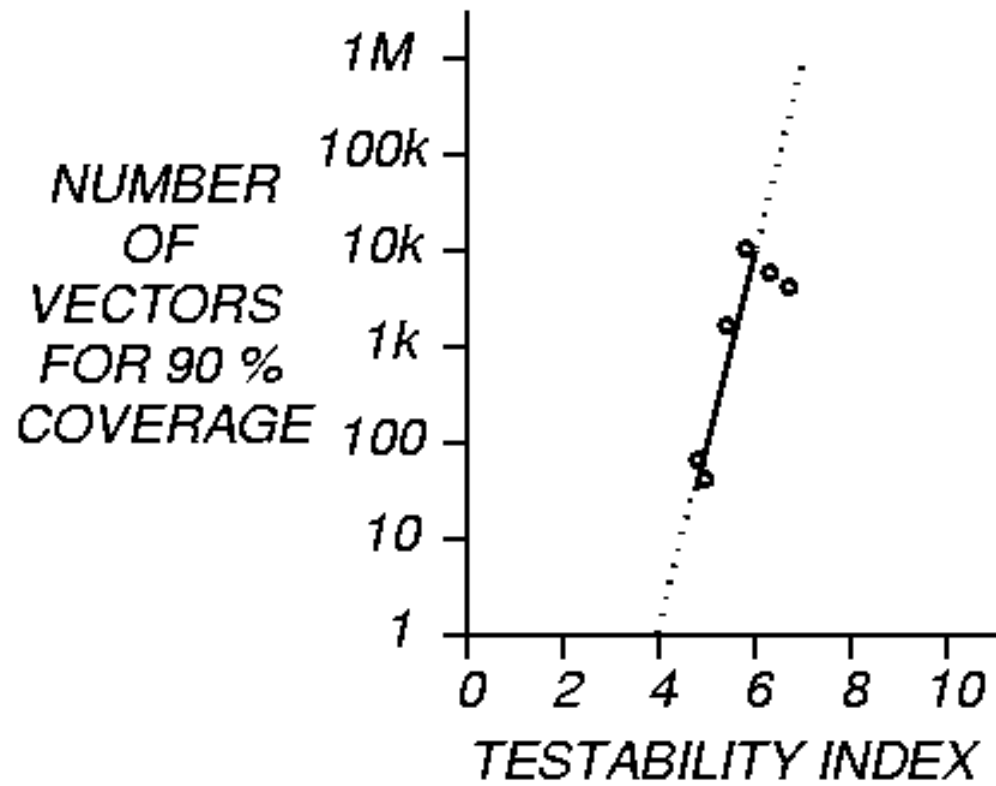




$R1 \rightarrow R1 : 0$
 $R2 \rightarrow R1 : 1$
 $R3 \rightarrow R1 : 2$
 $R4 \rightarrow R1 : 2$



$R1 \rightarrow R2 : 1$
 $R2 \rightarrow R2 : 0$
 $R3 \rightarrow R2 : 1$
 $R4 \rightarrow R2 : 1$



References

- **[Brglez 84] F. Brglez, P. Pownall and R. Hum, "Application of testability analysis: from ATPG to critical delay path tracing," Int'l Test Conf., 1984.**
- **[Goldstein1979] L. H. Goldstein E. L. Thigpen, SCOAP: Sandia controllability/observability analysis program, Design Automation conference , 1979.**